BE-202 · BTech First Year · UIT, Barkatullah University, Bhopal

Basic Electrical Engineering — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

UNIT I Introduction to Electrical Engg. & DC Circuits

1. Introduction: Generation, Transmission, Distribution & Utilization

Electrical energy ek chain mein user tak pahunchti hai: **Generation (11–33 kV) → Step-up → Transmission (220/400/765 kV) → Distribution (11 kV / 415 V) → Utilization (230 V)**, India mein 50 Hz.

- **Generation:** Heat, water, wind, sun, nuclear energy ko alternator se electrical energy mein badalna.
- **Transmission:** High voltage (HV/EHV) par bulk power door tak bhejna. High voltage kyun? P = VI fixed ho to V badhane par I kam → **I^2R loss kam**, conductor patla (sasta).
- **Distribution:** Substation se consumers tak: primary (11 kV feeders) aur secondary (415 V 3-ph / 230 V 1-ph).
- **Utilization:** Motors, lighting, heating, traction, electrolysis etc. mein use.

Conventional vs Renewable Energy Sources

Conventional (non-renewable)	Renewable (non-conventional)
Thermal (coal/steam, Rankine cycle), **Gas** turbine, **Diesel**, **Nuclear** (U-235 fission). **Large Hydro** bhi conventional maana jaata hai (though renewable).	**Solar** PV/thermal, **Wind**, **Small hydro**, **Biomass**/biogas, **Geothermal**, **Tidal**, wave, fuel cell.
Limited reserves, pollution (CO_2, SO_2), reliable & dispatchable, large capacity	Free & clean, inexhaustible, but intermittent, low capacity factor, storage chahiye

DER **Distributed Energy Resources** = chhote generating/storage units jo load ke paas (distribution level) lage hote hain: rooftop solar, small wind, micro-turbine, fuel cell, diesel genset, battery, EV. Fayde: T&D loss kam, reliability badhti hai, grid par load kam.

ENERGY STORAGE SYSTEMS Energy ko baad mein use ke liye store karna: **Battery** (Lead-acid, Li-ion), **Pumped hydro storage** (sabse bada), **Flywheel** (kinetic), **Supercapacitor**, **Compressed air (CAES)**, **SMES** (superconducting magnetic), Hydrogen/fuel cell. Use: renewable ki intermittency smooth karna, peak shaving, backup (UPS).

2. DC Circuits — Basics

Ohm's law: V = IR (temperature constant).

KCL: Kisi node par $\sum I = 0$ (aane wali current = jaane wali). Charge conservation.

KVL: Kisi closed loop mein $\sum V = 0$ ($\sum$ emf = $\sum$ IR drops). Energy conservation.

Series: $R_{eq} = R_1 + R_2 + \ldots$; **Parallel:** $1/R_{eq} = 1/R_1 + 1/R_2 + \ldots$

Maxwell's Loop (Mesh) Equations

Har independent loop mein ek **circulating (loop) current** assume karo (clockwise). Common branch mein net current = difference of loop currents. Har loop par KVL lagao → simultaneous equations. Number of equations = b − n + 1 (b = branches, n = nodes).

Matrix form: $[R][I] = [V]$; R_{11} = self-resistance of loop 1, R_{12} = −(common resistance).

Node (Nodal) Equations

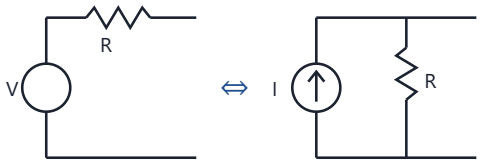
Ek node ko **reference (ground)** lo. Baaki (n−1) nodes ke voltages unknown. Har node par KCL: $\sum (V_{node} - V_{adjacent})/R$ = source current. Jab nodes kam aur loops zyada hon tab nodal better hai.

Source Conversion

Practical voltage source (V in series with R) ⇔ practical current source (I in parallel with same R).

$$I = V/R \quad \Leftrightarrow \quad V = I \cdot R$$

Ideal voltage source (R=0) aur ideal current source (R=∞) ka conversion possible nahi.



Voltage source ⇔ Current source

3. Network Theorems

(a) Superposition Theorem

Linear, bilateral network mein multiple sources hon to kisi branch ka response = har source **akele** lagane par mile responses ka **algebraic sum**. Ek source rakhte waqt baaki **voltage sources short** karo, **current sources open** karo (internal resistance rehne do). *Power par apply nahi hota* ($P \propto I^2$, non-linear).

(b) Thevenin's Theorem

Koi bhi linear two-terminal network = ek voltage source $\mathbf{V_{th}}$ series mein $\mathbf{R_{th}}$.

- $\mathbf{V_{th}}$ = load hata kar terminals A-B ka open-circuit voltage.
- $\mathbf{R_{th}}$ = sources ko replace (V short, I open) karke A-B se dekha gaya resistance.
- Load current: $I_L = V_{th}/(R_{th} + R_L)$

(c) Norton's Theorem

Network = current source $\mathbf{I_N}$ parallel mein $\mathbf{R_N}$. I_N = A-B short karke short-circuit current; $R_N = R_{th}$. Relation: $V_{th} = I_N R_N$; $I_L = I_N \cdot R_N/(R_N+R_L)$.

> **SOLVED EXAMPLE** 12 V source, 4 Ω series, phir 12 Ω shunt across A-B; R_L = 6 Ω. Thevenin: $V_{th} = 12\times12/(4+12) = 9$ V; $R_{th} = 4\|12 = 3$ Ω. $I_L = 9/(3+6) =$ **1 A**. Norton: $I_N = 12/4 = 3$ A, $R_N = 3$ Ω → $I_L = 3\times3/9 = 1$ A (same ✓). Max power: $R_L = 3$ Ω, $P_{max} = 9^2/(4\times3) =$ **6.75 W**.

(d) Maximum Power Transfer Theorem

DC network se load ko maximum power tab milegi jab $\mathbf{R_L = R_{th}}$. Is condition par efficiency sirf **50%** hoti hai.

$$P_{max} = \frac{V_{th}^{\,2}}{4R_{th}} \quad (\text{AC: } Z_L = Z_{th}^{*} \text{ i.e. complex conjugate})$$

Proof sketch: $P = V_{th}^2 R_L/(R_{th}+R_L)^2$; $dP/dR_L = 0 \Rightarrow R_L = R_{th}$.

(e) Millman's Theorem

Kai parallel voltage sources (har ek apne series R ke saath) ko ek equivalent source se replace kar sakte hain:

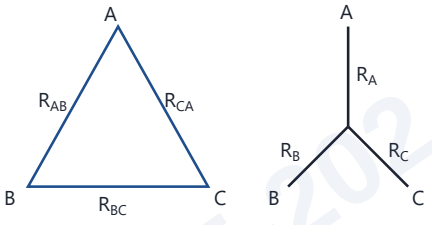
$$V_{eq} = \frac{V_1G_1 + V_2G_2 + \ldots + V_nG_n}{G_1 + G_2 + \ldots + G_n}, \quad R_{eq} = \frac{1}{G_1+G_2+\ldots+G_n}, \quad G = 1/R$$

Example: 10 V/2 Ω aur 6 V/2 Ω parallel → $V_{eq} = (5+3)/(0.5+0.5) = 8$ V, $R_{eq} = 1$ Ω.

(f) Reciprocity Theorem

Linear, bilateral, single-source network mein: agar branch-1 mein lagi source E se branch-2 mein current I aata hai, to wahi E branch-2 mein lagane par branch-1 mein same I aayega. Ratio **E/I = constant (transfer resistance)**. Dependent sources ke saath valid nahi.

(g) Star–Delta (Y–Δ) Transformation



Delta (left) ⇔ Star (right)

Δ → Y: $R_A = R_{AB}R_{CA} / (R_{AB}+R_{BC}+R_{CA})$ (do adjacent ka product / sum)

Y → Δ: $R_{AB} = R_A + R_B + R_AR_B/R_C$

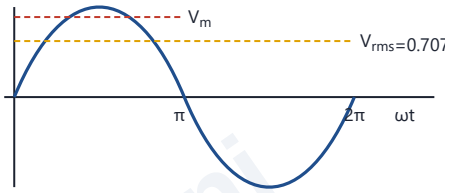
Balanced: $R_Y = R_\Delta/3$, $R_\Delta = 3R_Y$. Example: Δ ke teen 9 Ω → Y ke teen 3 Ω.

UNIT II Single Phase & Poly Phase AC Circuits

1. Alternating Waves & Values

Alternating quantity wo hai jiska magnitude time ke saath badalta hai aur direction periodically reverse hoti hai. Sinusoidal wave sabse common: **instantaneous value** $v = V_m \sin \omega t$, $\omega = 2\pi f$, $T = 1/f$. Waves: sinusoidal, square, triangular, sawtooth.

- **Instantaneous value:** kisi bhi instant par value (v, i).
- **Peak/Max value (V_m):** ek cycle ki highest value.
- **Average value:** half-cycle ka mean (full cycle ka sine average = 0). $V_{av} = 2V_m/\pi = 0.637V_m$
- **RMS (effective) value:** wo DC value jo same resistor mein same heat produce kare. $V_{rms} = V_m/\sqrt{2} = 0.707V_m$



v = V_m sin ωt

$$\text{Form factor } K_f = \frac{\text{RMS}}{\text{Average}} = 1.11 \text{ (sine)} ; \quad \text{Peak (crest) factor } K_p = \frac{\text{Max}}{\text{RMS}} = 1.414 \text{ (sine)}$$

Waveform	V_{av}	V_{rms}	K_f	K_p
Sine	$0.637V_m$	$0.707V_m$	1.11	1.414
Square	V_m	V_m	1	1
Triangular	$0.5V_m$	$V_m/\sqrt{3} = 0.577V_m$	1.155	1.732
Half-wave rectified sine	V_m/π	$V_m/2$	1.57	2

Example: v = 325 sin 314t → V_{rms} = 325/√2 ≈ 230 V, f = 314/2π = 50 Hz, V_{av} = 0.637×325 ≈ 207 V.

2. Phasor, Power & Power Factor

Phasor = rotating vector (anticlockwise, ω rad/s) jiski length RMS value aur angle phase batata hai. Phase difference φ: leading/lagging. j-operator: A = a + jb = |A|∠θ.

Element	Impedance	Phase of I w.r.t. V	Power
Pure R	R	In phase (φ=0)	P = VI, pf = 1
Pure L	jX_L, $X_L = 2\pi fL$	I lags V by 90°	P = 0 (pf = 0 lag)
Pure C	$-jX_C$, $X_C = 1/(2\pi fC)$	I leads V by 90°	P = 0 (pf = 0 lead)

$$\text{Active } P = VI\cos\varphi \text{ (W)} ; \text{ Reactive } Q = VI\sin\varphi \text{ (VAR)} ; \text{ Apparent } S = VI \text{ (VA)} ; S^2 = P^2 + Q^2$$
$$\text{Power factor} = \cos\varphi = P/S = R/Z$$

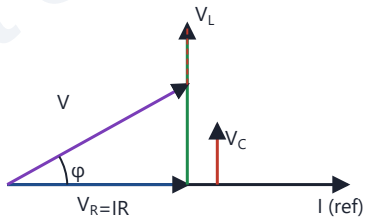
Low pf ke nuksaan: same power ke liye zyada current → zyada copper loss, bada conductor, poor voltage regulation. Improvement: shunt capacitor, synchronous condenser.

3. Single Phase Series & Parallel Circuits

Series RLC: $Z = R + j(X_L - X_C)$, $|Z| = \sqrt{(R^2 + (X_L - X_C)^2)}$, $\varphi = \tan^{-1}((X_L - X_C)/R)$. Current common hai, isliye **I ko reference** lo. V_R in phase, V_L 90° aage, V_C 90° peeche.

$X_L > X_C$ → inductive (lagging); $X_L < X_C$ → capacitive (leading).

Parallel circuit: Voltage common (reference). Admittance $Y = 1/Z = G - jB$ (S). Branch currents ka **phasor sum** = total current. $Y_{total} = Y_1 + Y_2 + \ldots$



Series RLC phasor ($X_L > X_C$, lagging)

SOLVED EXAMPLE R = 10 Ω, L = 0.1 H, C = 100 μF series, 230 V, 50 Hz. X_L = 31.4 Ω, X_C = 31.83 Ω → X = −0.43 Ω; Z ≈ 10.01 Ω, I ≈ 22.98 A, pf ≈ 0.999 leading (almost resonance), $P = I^2R$ ≈ 5.28 kW.

4. Resonance

Series resonance: $X_L = X_C \Rightarrow Z = R$ (minimum), I = V/R (**maximum**), pf = 1. Isliye **acceptor circuit**. Voltage magnification: $V_L = V_C = Q \cdot V$.

Parallel resonance (coil R-L ∥ C): Z_{dyn} = **L/(CR)** (maximum), line current **minimum**, pf = 1 → **rejector circuit**. Current magnification = Q.

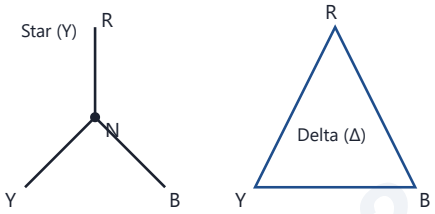
$$f_r = \frac{1}{2\pi}\sqrt{\left(\frac{1}{LC} - \frac{R^2}{L^2}\right)}$$

$$f_r = \frac{1}{2\pi\sqrt{(LC)}}\ ;\ \ Q = \frac{\omega_r L}{R} = \frac{1}{R}\sqrt{(L/C)}\ ;\ BW = f_2 - f_1 = R/(2\pi L) = f_r/Q$$

5. Poly Phase (Three Phase) AC Circuits

3-phase system mein teen equal emf, 120° apart. Fayde: same power ke liye kam conductor material, constant (non-pulsating) power, 3-ph induction motor self-starting, better efficiency.

Phase sequence: Order jisme phases apna maximum value lete hain: **R-Y-B** (positive). Reverse (R-B-Y) karne par 3-ph motor ki direction ulti ho jaati hai.



Star (3-ph 4-wire) and Delta (3-wire)

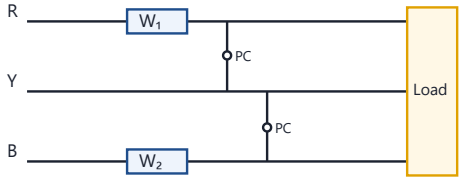
	Star (Y)	Delta (Δ)
Voltage	$V_L = \sqrt{3}\, V_{ph}$	$V_L = V_{ph}$
Current	$I_L = I_{ph}$	$I_L = \sqrt{3}\, I_{ph}$
Phase shift	V_L leads V_{ph} by 30°	I_L lags I_{ph} by 30°
Neutral	Available	Not available

$$\text{3-ph power: } P = 3V_{ph}I_{ph}\cos\varphi = \sqrt{3}\, V_L I_L \cos\varphi\ (W)\ ;\ Q = \sqrt{3}\, V_L I_L \sin\varphi\ ;\ S = \sqrt{3}\, V_L I_L$$

Example: 400 V, balanced star load, Z_{ph} = 8 + j6 Ω. V_{ph} = 400/√3 = 230.9 V, Z = 10 Ω, I_{ph} = I_L = 23.1 A, pf = 0.8 lag, P = √3×400×23.1×0.8 ≈ 12.8 kW.

Three Phase Power Measurement & Two Wattmeter Method

Methods: three-wattmeter (4-wire, unbalanced bhi), **two-wattmeter** (3-wire, balanced/unbalanced, star/delta), one-wattmeter (balanced only). **Blondel's theorem:** n-wire system ke liye (n−1) wattmeter kaafi hain.



CCs in R & B lines, PCs to Y

$$W_1 = V_L I_L \cos(30° - \varphi),\ \ W_2 = V_L I_L \cos(30° + \varphi)$$

$$P = W_1 + W_2\ ;\ \ \tan\varphi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

Q = √3(W₁ − W₂).

pf effect: pf = 1 → $W_1 = W_2$; pf = 0.5 → W_2 = 0; pf < 0.5 → W_2 negative (PC/CC reverse karke padho, subtract karo); pf = 0 → $W_1 = -W_2$, P = 0.

EXAMPLE W_1 = 5 kW, W_2 = 1 kW. P = 6 kW; tan φ = √3×4/6 = 1.155 ⇒ φ = 49.1°, pf = 0.655.

UNIT III Magnetic Circuits, Electrostatics & Transformer

1. Fundamental Definitions

Flux φ — Weber (Wb)

Flux density B = φ/A — Wb/m² = Tesla

Field intensity H = NI/l — AT/m

Magnetic Ohm's law: $\varphi = \frac{\text{MMF}}{\text{Reluctance}} = \frac{NI}{l/(\mu_0\mu_r A)}$

Ampere's circuital law: Closed path ke along H ka line integral = enclosed current: ∮H·dl = ∑I = NI. Isi se MMF calculate hota hai.

Faraday's law: Changing flux se emf induce: e = −N dφ/dt.

Permeability μ = $\mu_0\mu_r$ = B/H, $\mu_0 = 4\pi\times10^{-7}$ H/m

MMF F = NI — Ampere-turns (AT)

Reluctance S = l/($\mu_0\mu_r$A) — AT/Wb; Permeance = 1/S

Lenz's law: Induced emf/current ki direction aisi hoti hai ki wo apne cause (flux change) ko **oppose** kare (isliye minus sign).

Fleming's right hand rule (generator) — thumb motion, first finger field, middle finger induced current.

Calculation of MMF (series & composite magnetic circuit)

Har part (iron, air gap) ke liye: AT = H × l = (B/$\mu_0\mu_r$) × l. Total AT = ∑ AT of all parts (series circuit). Air gap mein μ_r = 1, isliye chhote gap ko bhi bahut AT chahiye.

EXAMPLE Iron ring: mean length 1 m, A = 10 cm^2, μ_r = 1000, air gap 1 mm, φ = 1 mWb. B = 1 T. Iron AT = $1\times1/(4\pi\times10^{-7}\times1000)$ = 796; Gap AT = $1\times0.001/(4\pi\times10^{-7})$ = 796. Total ≈ **1592 AT** (1 mm gap = 1 m iron!).

Comparison of Electric, Magnetic & Electrostatic Fields/Circuits

Electric circuit	Magnetic circuit	Electrostatic field
EMF (V)	MMF (AT)	Potential difference V
Current I = E/R	Flux φ = F/S	Electric flux ψ = Q (Coulomb)
Resistance R = ρl/A	Reluctance S = l/μA	Elastance 1/C
Conductivity σ	Permeability μ	Permittivity $\varepsilon = \varepsilon_0\varepsilon_r$
Current density J = I/A	Flux density B = φ/A	Flux density D = Q/A = εE
Electric intensity E = V/l	Field intensity H = NI/l	Field strength E = V/d

2. Induced EMFs & Inductance

Dynamically induced emf: Conductor field mein move kare (generator). $e = Blv\sin\theta$.

Statically induced emf: Conductor stationary, flux change ho:
(i) **Self-induced** — coil ka apna current change: e = −L di/dt.
(ii) **Mutually induced** — paas wali coil ke current change se (transformer): $e_2 = -M\,di_1/dt$.

INDUCTANCE

$$L = \frac{N\varphi}{I} = \frac{N^2}{S} = \frac{\mu_0\mu_r N^2 A}{l}\ (H)$$

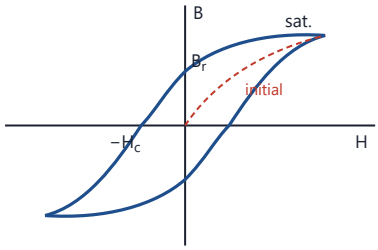
$$M = \frac{N_2\varphi_{12}}{I_1} = \frac{N_1N_2}{S};\quad M = k\sqrt{(L_1L_2)}$$

Energy stored = $\frac{1}{2}LI^2$; Series aiding: $L = L_1+L_2+2M$

Coefficient of coupling (k): $k = M/\sqrt{(L_1L_2)}$, $0 \le k \le 1$. k = 1 → tight coupling (saara flux dono coils link kare); k ≈ 0 → loosely coupled.

Leakage flux: Jo flux desired path (core/air gap) se bahar complete ho jaata hai. **Leakage coefficient** λ = total flux / useful flux (1.1–1.25). **Fringing:** Air gap mein flux lines bahar ki taraf phail jaati hain → effective gap area badh jaata hai, B kam ho jaata hai.

3. B-H Curve, Hysteresis & Eddy Current Loss



B-H curve & hysteresis loop

B-H (magnetisation) curve: Ferromagnetic material mein H badhane par B pehle tezi se badhta hai, phir **saturation** aa jaata hai (μ constant nahi).

Hysteresis: B ka H se peeche (lag) rehna.
Retentivity/Residual flux B_r = H=0 par bacha B. **Coercive force H_c** = B ko zero karne ke liye ulta H. Loop ka area = per cycle per m^3 energy loss.

Core loss: Hysteresis $P_h = \eta\, B_m^{1.6}\, f\, V$ (W) [Steinmetz] ; Eddy $P_e = K_e\, B_m^2\, f^2\, t^2\, V$ (W)

Eddy currents: Changing flux core mein khud circulating currents induce karta hai → I^2R heat. Kam karne ke liye: **laminated core** (thin sheets 0.35–0.5 mm, varnish insulated), high resistivity silicon steel. Hysteresis kam karne ke liye: **silicon steel / CRGO** (narrow loop). Soft magnetic (narrow loop) → transformer/machines; hard magnetic (wide loop) → permanent magnets.

4. Transformer

Transformer ek static device hai jo AC power ko ek circuit se doosre mein **mutual induction** se transfer karta hai, frequency same rehti hai, voltage/current level badalta hai.

Construction: Laminated CRGO silicon steel **core**; primary & secondary **windings** (copper/Al); insulation; tank, transformer oil (cooling + insulation), conservator, breather, bushings.

Principle: Primary par AC → core mein alternating flux → secondary se link → mutually induced emf (Faraday's law).

Types: (i) Core construction — **core type** (winding core ko gherti hai), **shell type** (core winding ko gherta hai), berry type; (ii) **step-up** ($N_2>N_1$) / **step-down**; (iii) single / three phase; (iv) use — power, distribution, instrument (CT/PT), auto, welding.

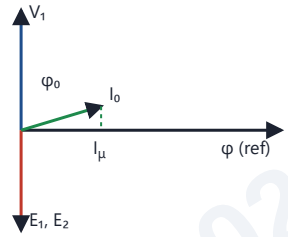
DC par kaam nahi karta (dφ/dt = 0, high current se jal sakta hai).

$$\text{EMF eq.: } E_1 = 4.44\, f N_1 \varphi_m;\quad E_2 = 4.44\, f N_2 \varphi_m;\quad K = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

Derivation: $\varphi = \varphi_m \sin \omega t$; average $d\varphi/dt$ (quarter cycle) = $\varphi_m/(1/4f) = 4f\varphi_m \rightarrow E_{av}$/turn = $4f\varphi_m$; × form factor 1.11 → E_{rms}/turn = $4.44\, f\varphi_m$.

Example: 2200/220 V, 50 Hz, φ_m = 0.01 Wb → $N_1 = 2200/(4.44\times50\times0.01) \approx 991$, $N_2 \approx 99$.

Phasor Diagram — No Load & On Load

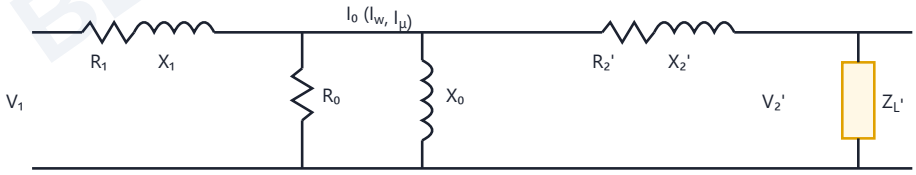


No-load phasor (ideal: $\varphi_0 \approx 80$–$85°$)

No load: I_0 = 2–5% of rated; I_0 ke do components: **$I_\mu = I_0 \sin \varphi_0$** (magnetising, φ ke in-phase) aur **$I_w = I_0 \cos \varphi_0$** (core-loss/working, V_1 ke in-phase). E_1, E_2 flux se 90° peeche. No-load pf bahut low (0.1–0.2).

On load: I_2 flow karta hai → secondary MMF N_2I_2 flux ko oppose karta hai → primary extra current **$I_2' = K I_2$** leta hai taaki flux constant rahe. $I_1 = I_0 + I_2'$ (phasor). Practical: $V_2 = E_2 - I_2(R_2 + jX_2)$, $V_1 = -E_1 + I_1(R_1 + jX_1)$.

Equivalent Circuit



Exact equivalent circuit referred to primary: $R_2' = R_2/K^2$, $X_2' = X_2/K^2$, $R_0 = V_1/I_w$, $X_0 = V_1/I_\mu$

Approximate circuit: shunt branch ($R_0 \| X_0$) ko input terminals par shift → $R_{01} = R_1 + R_2/K^2$, $X_{01} = X_1 + X_2/K^2$, $Z_{01} = \sqrt{(R_{01}^2 + X_{01}^2)}$.

Losses, Efficiency & All-day Efficiency

- **Iron/core loss P_i** (hysteresis + eddy) — constant loss (har load par same).
- **Copper loss $P_{cu} = I_1^2 R_{01}$** — variable loss, ∝ (load)². x fraction load par = $x^2 P_{cu,FL}$.

EFFICIENCY

$$\eta = \frac{x \cdot S \cdot \cos\varphi}{x \cdot S \cdot \cos\varphi + P_i + x^2 P_{cu,FL}} \times 100\%;\quad \eta_{max} \text{ when } P_{cu} = P_i \Rightarrow x = \sqrt{(P_i/P_{cu,FL})}$$

All-day (energy) efficiency = output kWh in 24 h / input kWh in 24 h. Distribution transformers ke liye important hai kyunki wo 24 h energised rehte hain (iron loss 24 h) par load variable hota hai. Isliye inhe low iron loss ke saath design karte hain.

EXAMPLE 20 kVA, P_i = 200 W, $P_{cu,FL}$ = 400 W, pf 0.8. Full load: η = 16000/(16000+600) = **96.4%**. Half load: output 8000 W, losses 200 + 100 = 300 → η = 96.4%. η_{max} at x = √(0.5) = 0.707 load.

Voltage Regulation

$$\%\,\text{Reg} = \frac{V_{2,NL} - V_{2,FL}}{V_{2,NL}} \times 100 \approx \frac{I_2(R_{02}\cos\varphi \pm X_{02}\sin\varphi)}{V_2} \times 100 \quad (+\text{lagging}, -\text{leading})$$

Leading pf par regulation negative bhi ho sakta hai (load par voltage badh jaata hai). Kam regulation achha.

OC and SC Test

	Open Circuit (OC) test	Short Circuit (SC) test
Side	LV side par rated V, HV open	HV side par reduced V (5–10%), LV shorted
Readings	V_o, I_o, W_o	V_{sc}, I_{sc} (= rated), W_{sc}
Gives	**Iron loss** = W_o; $R_0 = V_o/I_w$, $X_0 = V_o/I_\mu$, $\cos\varphi_0 = W_o/(V_oI_o)$	**Full-load Cu loss** = W_{sc}; $R_{eq} = W_{sc}/I_{sc}^2$, $Z_{eq} = V_{sc}/I_{sc}$, $X_{eq} = \sqrt{(Z^2 - R^2)}$

Welding, Auto & Three-phase Transformer

- **Welding transformer:** Step-down, low voltage (≈ 15–80 V open circuit) aur very high current (100s of A); high leakage reactance (drooping characteristic) taaki arc stable rahe aur short-circuit current limit ho. Current control: tapped reactor / movable core / magnetic shunt.
- **Auto transformer:** Sirf **ek winding**, part common to primary & secondary. Power partly conduction se, partly induction se transfer. Copper saving = K × (two-winding copper) jahan K = V_{LV}/V_{HV}. Fayde: sasta, chhota, better η & regulation. Nuksaan: no electrical isolation. Use: variac, motor starting, interconnecting grids.
- **Three-phase transformer:** 3 single-phase units ka bank ya ek common core (core/shell type) par 3 primary + 3 secondary. Connections: Y-Y, Δ-Δ, Y-Δ (step-down at substation), Δ-Y (step-up at generating station, distribution). Single 3-ph unit sasta aur chhota hota hai.

UNIT IV DC Machines & AC Machines

1. DC Generator

Principle: Faraday's law of electromagnetic induction — conductor magnetic field mein rotate kare to **dynamically induced emf** (e = Blv). Direction: Fleming's right hand rule. Armature mein emf AC hoti hai; **commutator** use DC mein badalta hai.

Construction: Yoke (frame, cast iron/steel, flux path), **pole core & pole shoe** (flux spread), **field winding**, **armature core** (laminated silicon steel, slots), **armature winding** (lap: A = P; wave: A = 2), **commutator** (copper segments, mica insulated), **brushes** (carbon), bearings, shaft.

$$\text{EMF eq.: } E_g = \frac{\varphi\, Z\, N\, P}{60\, A} \text{ volts}$$

φ = flux/pole (Wb), Z = total conductors, N = rpm, P = poles, A = parallel paths (lap A = P, wave A = 2)

Derivation: Ek conductor ek revolution mein Pφ flux cut karta hai, time 60/N s → emf/conductor = PφN/60; series conductors = Z/A.

Types

Type	Field connection	Relations
Separately excited	Field alag DC source se	$E_g = V + I_aR_a$
Shunt	Field armature ke parallel	$I_a = I_L + I_{sh}$, $I_{sh} = V/R_{sh}$
Series	Field armature ke series	$I_a = I_{se} = I_L$, $E_g = V + I_a(R_a+R_{se})$
Compound	Shunt + series (long/short shunt; cumulative/differential)	—

Magnetic (Open-circuit) characteristic: E_0 vs I_f at constant speed — B-H curve jaisa: shuru mein residual magnetism se small emf, phir linear, phir saturation. Self-excited generator build-up ke liye: residual flux hona chahiye, field resistance < **critical resistance**, field connection sahi (flux aid kare).

2. DC Motor

Principle: Current-carrying conductor magnetic field mein rakhne par force F = BIl lagta hai (Fleming's **left hand rule**) → torque → rotation.

Back EMF: Rotating armature mein bhi emf induce hoti hai jo supply voltage ko oppose karti hai (Lenz's law):

$$E_b = \frac{\varphi Z N P}{60A}\ ;\quad V = E_b + I_aR_a$$

E_b current ko self-regulate karti hai; start par $E_b = 0$ → bahut high current, isliye starter chahiye. Speed $N \propto E_b/\varphi$.

TORQUE

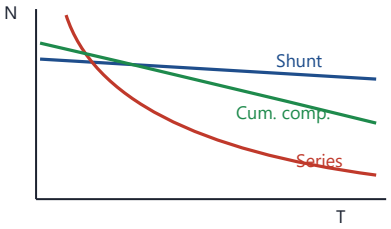
$$T_a = 0.159\, \varphi\, Z\, I_a\, \frac{P}{A} \text{ N·m}$$

$$T_a \propto \varphi I_a\ ;\quad E_bI_a = T_a \cdot \frac{2\pi N}{60}$$

$$T = 9.55\, E_bI_a/N$$

EXAMPLE 220 V shunt motor, R_a = 0.5 Ω, I_a = 20 A, 1000 rpm. E_b = 220 − 10 = 210 V. T = 9.55×210×20/1000 ≈ **40.1 N·m**. Agar I_a = 40 A (same φ): E_b = 200 V → N = 1000×200/210 ≈ 952 rpm.

Mechanical (Speed–Torque) Characteristics



N–T characteristics of DC motors

- **Shunt motor:** φ ≈ constant → $T \propto I_a$; speed almost constant (5–10% droop). **Constant speed motor.**
- **Series motor:** $\varphi \propto I_a$ (before saturation) → $T \propto I_a^2$; $N \propto 1/I_a$. High starting torque; **no-load par kabhi start mat karo** (speed dangerously high).
- **Compound:** beech ki characteristics.

Armature Reaction, Commutation & Interpoles

- **Armature reaction:** Armature current ka flux main field flux ko affect karta hai: (i) **cross-magnetising** — field distort hota hai, MNA (magnetic neutral axis) shift (generator mein rotation ki direction mein, motor mein opposite); (ii) **demagnetising** — main flux kam (saturation ki wajah se). Remedy: compensating winding (pole face mein), interpoles, high reluctance pole tips.
- **Commutation:** Coil jab brush ke neeche short-circuit hoti hai, uska current +I se −I reverse hona. Coil ki self-inductance (reactance voltage) reversal ko delay karti hai → **sparking**. Ideal = linear commutation. **Improvement:** resistance commutation (carbon brushes), **EMF commutation** — brush shift ya **interpoles**.
- **Interpoles (commutating poles):** Chhote poles main poles ke beech (GNA par), winding armature ke series mein. Ye reactance voltage ko neutralise karte hain aur cross-magnetisation cancel karte hain. Polarity: generator mein next main pole (rotation ki direction mein) jaisi, motor mein previous main pole jaisi.

Starting & Speed Control

Starting: Start par $E_b = 0 \rightarrow I_a = V/R_a$ (bahut zyada, e.g. 220/0.5 = 440 A). Isliye series resistance wala **starter**: **3-point starter** (shunt motor; NVC — no-volt coil, OLR — overload release), **4-point starter** (NVC supply ke across, field control ke liye better), 2-point (series motor). **Speed:** $N \propto \frac{E_b}{\varphi} = \frac{V - I_a R_a}{\varphi}$

Method	How	Speed range
Flux (field) control	Shunt field mein rheostat (φ↓ → N↑); series motor mein diverter	Above rated
Armature (rheostatic) control	Armature ke series resistance (I_aR drop ↑ → N↓); lossy	Below rated
Voltage control	Variable supply voltage — Ward-Leonard system, chopper	Both, smooth

Applications: Shunt — lathe, centrifugal pump, fans, blowers, conveyors, machine tools (constant speed). **Series** — electric traction, cranes, hoists, elevators (high starting torque). **Cumulative compound** — rolling mills, presses, shears, punches (sudden heavy loads).

3. AC Machines — Three Phase Induction Motor

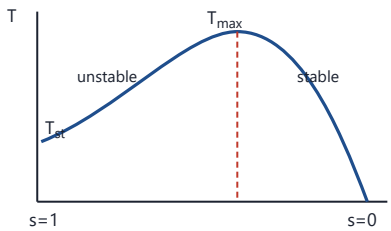
Construction: Stator — laminated core, 3-ph distributed winding. **Rotor** — (i) **squirrel cage** (Al/Cu bars short-circuited by end rings; simple, rugged, cheap) (ii) **slip-ring / wound rotor** (3-ph winding, slip rings + external resistance → high starting torque, speed control).

Principle: 3-ph supply stator mein **rotating magnetic field (RMF)** banata hai jo synchronous speed N_s se ghoomta hai. RMF rotor conductors ko cut karta hai → emf → current (rotor shorted) → force → rotor RMF ki direction mein ghoomta hai (Lenz's law: relative motion kam karne ki koshish). Rotor kabhi N_s tak nahi pahunchta, isliye **asynchronous motor**.

$$N_s = \frac{120\,f}{P} \;;\; s = \frac{N_s - N}{N_s} \;;\; f_r = sf$$

$$E_{2r} = sE_2 \;;\; X_{2r} = sX_2 \;;\; P_g : P_{cu2} : P_{mech} = 1 : s : (1-s)$$

$$E_{ph} = 4.44\, f\, N\, \varphi\, K_w$$



Torque–slip characteristic (N → right)

Torque-slip: Low slip (normal running) par $T \propto s$ (linear, stable); high slip par $T \propto 1/s$. T_{max} jab **$R_2 = sX_2$** ($s_m = R_2/X_2$); T_{max} R_2 par depend nahi karta. Full-load slip 2–5%. Rotor resistance badhane se starting torque badhta hai (slip-ring motor).

Example: 4-pole, 50 Hz, 1440 rpm → N_s = 1500 rpm, s = 60/1500 = 0.04, rotor frequency = 0.04×50 = 2 Hz.

Types & Applications: Squirrel cage — fans, pumps, lathes, compressors, conveyors (industry ka workhorse). Slip-ring — cranes, hoists, lifts, elevators. **Single-phase IM** self-starting nahi hai (pulsating field); split-phase / capacitor start → fans, washing machines, refrigerators.

4. Synchronous Machine

Construction: Stator = armature (3-ph winding). **Rotor** = field (DC excitation via slip rings/brushless exciter). Rotor types: **salient pole** (projecting poles, large diameter, short length, low speed — hydro, 4–60+ poles) aur **non-salient / cylindrical** (smooth, small diameter, long, high speed 1500/3000 rpm — steam/gas turbo-alternators, 2 or 4 poles).

Alternator principle: DC excited rotor ko prime mover se ghumate hain → rotating flux stator conductors ko cut karta hai → 3-ph emf (Faraday's law). Stationary armature: HV insulation easy, sirf 2 slip rings (low DC).

$$\text{EMF eq.: } E_{ph} = 4.44\, K_c\, K_d\, f\, \varphi\, T_{ph}\ ;\ \ f = \frac{PN_s}{120}\ ;\ \ V = E - I(R_a + jX_s)$$

K_c = pitch (coil-span) factor = cos(α/2), K_d = distribution factor, T_{ph} = turns/phase

Synchronous motor: Stator par 3-ph supply (RMF at N_s) + rotor par DC → rotor poles RMF se "lock" ho jaate hain aur exactly $\mathbf{N_s}$ par ghoomte hain. **Self-starting nahi** (damper winding se induction motor ki tarah start). Speed load se independent.

Characteristics: Alternator — terminal voltage vs load (lagging pf par voltage girta hai, leading par badhta hai). Sync motor — **V-curves** (I_a vs I_f): under-excited → lagging pf, over-excited → leading pf. Isliye over-excited sync motor ko **synchronous condenser** (pf improvement) ki tarah use karte hain.

Applications: Alternators — saare power stations. Sync motor — constant speed drives (compressors, rolling mills, cement mills), pf correction, clocks.

UNIT V Power Systems & Protective Devices

1. Conventional Power Systems & Present-day Scenario

Conventional power system: Bade centralised generating stations (thermal, hydro, nuclear) → long HV transmission → distribution → consumers. Power flow **one-way**, passive consumers, limited monitoring.

Present-day scenario (India): Installed capacity ~450 GW+; coal-based thermal abhi bhi majority generation deta hai, par renewable (solar + wind + hydro) ka share tezi se badh raha hai (target 500 GW non-fossil by 2030). "One Nation One Grid" — saare regional grids synchronised (National Grid). HVDC links, 765 kV AC, rooftop solar, EVs, smart meters, deregulation (GENCO/TRANSCO/DISCOM). Challenges: AT&C losses, RE intermittency.

2. Structure of Power System



Power system structure (large industries 33–132 kV, commercial 11 kV, homes 230 V)

Bulk Power Grid	Micro-grid	Smart Grid
Big generating stations + EHV transmission ka interconnected network (e.g., India's National Grid; 5 regional grids). Fayde: reliability, reserve sharing, economic dispatch.	Local, small-scale grid (DERs + storage + loads) jo main grid ke saath **grid-connected** ya **islanded** mode mein chal sakta hai. E.g., campus, village solar micro-grid. Fayde: reliability, remote electrification, low losses.	Conventional grid + **two-way communication**, sensors, automation & IT. Features: smart meters (AMI), two-way power flow (prosumers), demand response, self-healing, PMU/SCADA monitoring, renewable & EV integration.

3. Types of Load & Load Characteristics

Domestic (lights, fans, AC; evening peak), **Commercial** (shops, offices; day-time), **Industrial** (nearly constant, high LF), **Agricultural** (pumps; seasonal), **Municipal** (street lighting, water supply), **Traction** (railways, metro; rush-hour peaks).

Load curve: Load (kW) vs time (24 h) graph; area = energy consumed (kWh). **Load duration curve** = loads descending order mein. By nature: resistive, inductive (motors), capacitive; constant power/current/impedance loads.

$$\text{Demand factor} = \frac{\text{Max demand}}{\text{Connected load}} \ (\leq 1) \ ; \quad \text{Load factor} = \frac{\text{Average load}}{\text{Max demand}} \ (\leq 1)$$

$$\text{Diversity factor} = \frac{\sum \text{individual max demands}}{\text{Max demand of system}} \ (\geq 1) \ ; \quad \text{Plant capacity factor} = \frac{\text{Actual energy produced}}{\text{Max possible energy (capacity} \times \text{time)}}$$

EXAMPLE Max demand 20 MW, annual energy 87.6×10^6 kWh. Avg load = $87.6\times10^6/8760$ = 10 MW → LF = 10/20 = **0.5**. High load factor = plant better utilised, per unit cost kam.

4. Fault Analysis

Fault = abnormal condition jab insulation fail ho ya conductors aapas mein / ground se touch ho jaayen → very high current, low voltage, equipment damage, instability.

Fault type	Occurrence	Severity
Symmetrical — LLL / LLLG (teeno phases short, system balanced rehta hai)	~5% (rarest)	Most severe
Unsymmetrical — LG (single line to ground)	~70–80% (most common)	Least severe
LL (line to line) / LLG (double line to ground)	~15% / ~10%	Medium / medium-high

Analysis: Symmetrical faults → **per-unit system** + Thevenin equivalent: $I_f = V_{th}/Z_{th}$; **fault MVA** = base MVA / Z_{pu}. Unsymmetrical faults → **symmetrical components** (positive, negative, zero sequence; Fortescue). Uses: CB rating, relay setting. Open-circuit (series) faults bhi hote hain.

Example: 10 MVA generator, X = 0.1 pu → 3-ph fault MVA = 10/0.1 = 100 MVA.

5. Protective Devices — Relays

Requirement of relays: Relay fault ko **sense** karta hai (CT/PT ke through) aur circuit breaker ki trip coil energise karta hai taaki faulty section isolate ho jaaye, baaki system chalta rahe. Chain: **Fault → CT/PT → Relay → Trip coil → CB opens**.

Qualities: Selectivity (sirf faulty part disconnect), **Speed** (fast), **Sensitivity** (chhote fault par bhi operate), **Reliability**, **Stability** (external fault par mal-operate na kare), simplicity & economy.

Relay types: Electromagnetic attraction & induction, thermal, static, numerical (microprocessor). By function: overcurrent, distance (impedance), differential (transformer/generator), directional, Buchholz (oil-filled transformer).

6. Circuit Breakers & Operating Principle

Circuit breaker = switching device jo normal aur fault dono conditions mein circuit ko make/break kar sake (manually ya relay se automatically). **Operating principle:** CB mein fixed aur moving contacts hote hain. Fault par relay trip coil energise karta hai → mechanism (spring/pneumatic) contacts alag karta hai → contacts ke beech **arc** banta hai → arc ko medium (oil/air/SF_6/vacuum) mein cool/lengthen/deionise karke current zero par extinguish karte hain. Arc extinction: **high resistance method** (DC/low power) aur **low resistance / current-zero method** (AC CBs). Terms: restriking voltage, recovery voltage, breaking capacity.

CB type	Arc quenching medium	Voltage range / use
Air break / ACB, MCB, MCCB	Atmospheric air, arc chutes	LV (up to ~1 kV); MCB for domestic
Oil CB (bulk / minimum oil)	Transformer oil (H_2 gas cools arc)	Up to 132 kV (old)
Air blast CB	High pressure compressed air	132–400 kV
SF_6 CB	Sulphur hexafluoride (electronegative, high dielectric strength)	33–765 kV (most popular EHV)
Vacuum CB	Vacuum (10^{-5} torr); no medium to ionise	Up to 33 kV (MV), low maintenance

Fuse vs CB: Fuse sasta, one-time device, khud sense+interrupt karta hai; CB reusable, relay ke saath remote/auto operation.

7. Types of Insulators

Overhead line conductors ko towers/poles se insulate aur support karne ke liye. Material: **porcelain** (most common), toughened **glass**, **polymer/composite** (silicone rubber, light weight).

- **Pin type:** cross-arm par pin se fixed; up to 33 kV (above it uneconomical).
- **Suspension (disc):** discs ki string (~11 kV/disc), conductor neeche latakta hai; above 33 kV; ek disc fail → sirf wahi replace. Conductor ke paas wali disc par max voltage; **string efficiency** = $V_{string}/(n \times V_{nearest\ disc})$, guard ring se improve.
- **Strain type:** horizontal disc string; high tension — dead end, sharp turns, river crossings.
- **Shackle:** spool shape, pole par bolt; LV lines.
- **Post type:** base par mounted; substation bus-bars.
- **Stay (egg) insulator:** stay wire mein; LV poles.